

Linux Deep Dive

Daniel Esteban Vela López

SPTI+

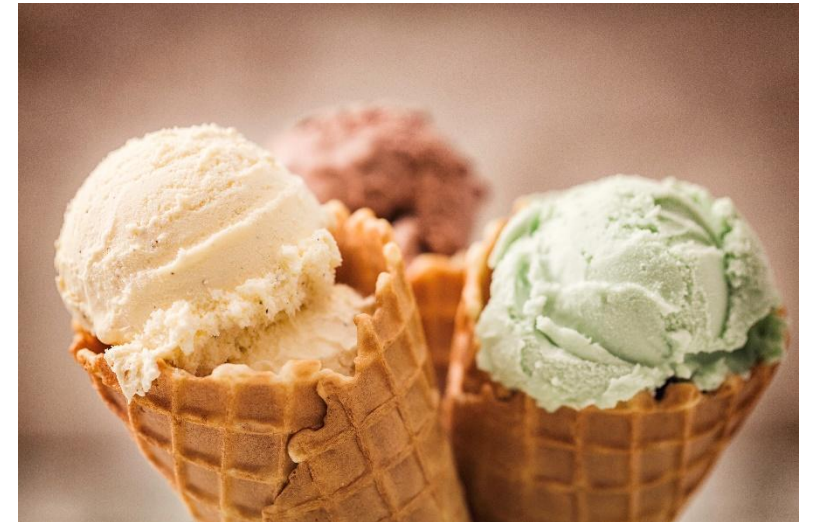
What is Linux?

- Open source Operative System (OS)
- OS is a resource manager
 - Processor time
 - Memory
 - Storage space
- Kernel
 - OS core
 - OS's lowest level



What is a Linux distro?

- Desktop environment
 - Package manager
 - Apps
 - Objectives and philosophy
-
- *Think of Linux Distros as ice cream flavors.*





Why use Linux in cybersecurity?

- Linux is the most used operating system by cybersecurity professionals due to its control, flexibility, and its great tool ecosystem.
- Kali Linux is a specialized distro based on Debian that includes tools for pentesting, forensics, network analysis, reverse engineering, etc.
- Working with the terminal (CLI) directly allows for task automation, direct interaction with tools and a better understanding of how systems work.

Linux Filesystem Structure

Directory	Typical contents
/	System root
/home	Users' personal directories
/etc	System configuration files
/bin	Essential system commands
/usr	User applications
/var	Variable data (logs, cache)
/tmp	Temporary files
/root	root user's home directory
/dev	Devices (disks, USB, etc.)
/proc	Information about processes and the kernel

Essential navigation and manipulation commands

Navigation and exploration



```
pwd          # Show the current directory
ls           # List files
cd           # Change directory
```


File and directory manipulation



```
cp file destination      # Copy
mv file destination     # Move or rename
rm file                 # Delete
mkdir directory_name    # Create directory
touch file.txt          # Create an empty file
cat file.txt            # See contents of a file
less file.txt           # See paginated contents of a file
```

Useful exploration



```
tree                # See directory structure
file file           # Detect file type
find /path -name "*.txt" # Find files (in this case, by extension/name)
```

File permissions and user management

Understanding permissions



```
ls -l
```

```
# Typical output
```

```
-rwxr-xr-- 1 user group 1234 jul 18 file.sh
```

The first 10 characters indicate type and permissions:

- r = read
- w = write
- x = execute

Structure: [user][group][others]

Permissions and user commands



```
chmod 755 file.sh      # Change permissions (rwxr-xr-x)
chown user:group file  # Change owner
whoami                 # Show current user
sudo command           # Execute with privileged permissions
```

Processes and services

Process monitoring



```
ps aux          # See all processes
top / htop      # Dynamic view (htop must be installed)
kill PID        # End process
```


Services (systemd)



```
systemctl status service_name  
systemctl start/stop/restart service_name
```

Package installation and use of man

Package installation



```
sudo apt update           # Update the packages list
sudo apt install name     # Install a package
sudo apt remove name      # Remove a package
```

Get help



```
man command      # Complete manual  
command --help   # Quick help
```

Basic concepts of Bash scripting

Basic concepts of Bash scripting

Scripts can automate tasks using command tools.



```
#!/bin/bash  
echo "Hello, World"
```

Useful script commands

- `if, then, else`
- `for, while`
- `read` (read user input)
- `$(command)` (command substitution)

Simple example



```
#!/bin/bash  
echo "What is your name?"  
read name  
echo "Hello, $name"
```


Allow execution



```
chmod +x script.sh  
./script.sh
```



Gracias